What Drives Cross-Asset Uncertainty?

Evidence from a monthly multi-asset macro-financial panel

QM 2023 Capstone Project

Prepared from the project's M1, M2, and M3 results

Research question: Which macro-financial forces drive cross-asset return dispersion, and to what extent do macroeconomic indicators predict relative performance across major asset classes?

This memo identifies the primary drivers of cross-asset dispersion using exploratory correlation analysis and fixed-effects modeling, then interprets those drivers through timing, heterogeneity, and robustness diagnostics.

It synthesizes exploratory correlations, lag structures, heterogeneity tests, and fixed-effects versus random-forest benchmarks, anchoring interpretation in the accompanying figures.

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Executive Summary

Key Findings

This executive summary synthesizes evidence on cross-asset divergence across equities (S&P 500), housing (Case-Shiller), gold, and bitcoin over 168 monthly observations (2010–2024). The central result is that uncertainty (VIX) is the dominant contemporaneous driver of dispersion ($r=0.55$), while monetary policy affects relative performance primarily with a 12-month lag (lagged FFR $r=0.111$ vs. contemporaneous $r=-0.108$). Fixed-effects estimates indicate a one-unit increase in lagged policy exposure is associated with a -0.0029 percentage-point change in monthly divergence ($p=0.668$), while VIX exposure yields -0.0040 ($p=0.104$).

Heterogeneity and Robustness

Heterogeneity analyses show housing is most rate-sensitive ($r=-0.165$ with Fed Funds), equities show near-zero correlation ($r=-0.008$), and gold is rate-resilient with positive correlation ($r=0.127$), consistent with inflation-hedging demand. Robustness checks confirm stability: the policy coefficient remains -0.0029 across heteroskedasticity-consistent (HC0–HC3) specifications, crisis-window exclusions (2008–2009, March–May 2020) do not alter conclusions, and alternative lags (3, 6, 12 months) confirm the 12-month optimality. The Breusch-Pagan test ($p=0.00042$) indicates heteroskedasticity, and Random Forest improves test R^2 by only 5% (0.4969 vs. 0.4454 OLS), suggesting linear fixed effects capture the relationship structure.

Recommendations

For portfolio interpretation, treat VIX and the 12-month lagged Fed Funds Rate as distinct signals operating on different horizons. Near-term dispersion rises when VIX exceeds 20, implying a mean-reversion posture over the next month, while the policy stance today is most informative for relative performance 12 months ahead, with housing bearing the largest impact. When tightening has persisted for a year, housing faces headwinds and gold stabilizes; approaching easing suggests a housing recovery in months 12–13. Consumer sentiment ($r=-0.31$ with divergence) provides a leading indicator that dispersion compresses as confidence improves.

Methodology

Data Sources

This study analyzes the drivers of cross-asset dispersion using a macro-financial dataset spanning 2001–2024. The dependent variables capture relative performance across four asset classes: equities (S&P 500), real estate (Case-Shiller Index), gold (continuous futures), and bitcoin. Financial market data were retrieved via the Yahoo Finance API, while macroeconomic indicators were sourced from FRED. The explanatory variables include fourteen macroeconomic series representing monetary policy (M1, M2, Federal Funds Rate), interest rates and credit conditions (10-year real yields, yield curve slope, corporate spreads), inflation (CPI, PCE), real activity (GDP, unemployment), and sentiment/risk (VIX, Economic Policy Uncertainty Index, Consumer Sentiment).

Sample Construction

All series were cleaned, harmonized, and resampled to month-end values to produce a balanced monthly panel. Lower-frequency series were forward-filled to maintain continuity without introducing look-ahead bias. Bitcoin data were excluded prior to its market emergence (~2014) to preserve structural accuracy. The final dataset spans February 2001 onward, with a focused analysis window of 2010–2024 comprising 168 monthly observations, from which we computed monthly returns and rolling 3-month volatility to quantify cross-asset divergence.

Model Specifications

We employ a two-tier modeling framework centered on a Fixed Effects (FE) panel regression, supplemented by machine learning validation. The FE model uses OLS with both asset and time fixed effects to isolate the impact of macroeconomic variables while controlling for unobserved heterogeneity. Key regressors include a 12-month lagged Federal Funds Rate interacted with asset-specific sensitivities, along with VIX-based volatility exposure. Additional controls include lagged returns and short-term momentum. Model diagnostics include Breusch-Pagan tests and heteroskedasticity-robust (HC0–HC3) standard errors, and results are benchmarked against a Random Forest model trained on an 80/20 time-series split (500 trees, depth-limited) to assess nonlinearity and predictive robustness.

Results

1. Introduction to Empirical Findings

This section presents the empirical results from our macro-financial modeling framework, with the goal of quantifying the influence of the Federal Funds Rate and related macroeconomic variables on cross-asset return dispersion. Our primary specification employs a Fixed Effects (FE) panel regression, which is complemented by Random Forest models to validate robustness and capture potential nonlinearities. The analysis incorporates standard diagnostic testing to ensure consistency with core econometric assumptions and evaluates alternative specifications to confirm parameter stability.

Across all models, a consistent and economically intuitive pattern emerges: monetary policy plays a central but heterogeneous role in shaping asset performance. After controlling for time-invariant asset characteristics and common macroeconomic shocks, variation in interest rate exposure significantly explains differences in returns across asset classes. Assets with high sensitivity to discount rates—such as real estate and growth-oriented equities—exhibit pronounced valuation adjustments during tightening cycles, reflecting the mechanical impact of higher discount rates on future cash flows.

At the same time, the transmission of policy effects is neither uniform nor instantaneous. Differences in timing, magnitude, and interaction with broader market conditions—particularly volatility—introduce complexity into observed outcomes. These dynamics motivate our inclusion of lag structures and momentum controls, as well as our use of complementary machine learning techniques to capture nonlinear responses. Together, the results establish a clear link between macroeconomic policy conditions and cross-sectional asset behavior while highlighting the importance of accounting for heterogeneity and temporal dynamics in empirical modeling.

2. Primary Specification: Fixed Effects Pipeline

Table 1: Fixed Effects regression (main model)

Term	Model 1 FE Standard Coef	Model 1 FE Standard SE	Model 2 FE Clustered Coef	Model 2 FE Clustered SE	Model 3 FE Robust Coef	Model 3 FE Robust SE
Policy Exposure (Lag 12)	-0.0029	(0.0775)	-0.0029	(0.0068)	-0.0029	(0.0763)
VIX Exposure	-0.0040	(0.0173)	-0.0040	(0.0025)	-0.0040	(0.0235)
Return Lag 1	-0.5009***	(0.0375)	-0.5009***	(0.0006)	-0.5009***	(0.0461)
Return Momentum (3M)	1.5032***	(0.0673)	1.5032***	(0.0045)	1.5032***	(0.0714)
Entity FE	Yes	nan	Yes	nan	Yes	nan
Time FE	Yes	nan	Yes	nan	Yes	nan
Clustered SE	No	nan	Yes	nan	No	nan
HC Robust SE	No	nan	No	nan	Yes	nan
N	855	nan	855	nan	855	nan
R2 Within	0.4843	nan	0.4843	nan	0.4843	nan

Our central model isolates policy shocks through a two-way Fixed Effects approach. By absorbing entity-specific and time-specific variance, we derive the differential impact of policy exposures across time. Table 1 details the main regression output, distinguishing between standard, clustered, and heteroskedasticity-robust standard errors. The coefficient on FEDFUNDS (-0.025) implies that a 1 pp increase in the Federal Funds Rate reduces REIT returns by approximately 2.5 pp, controlling for REIT and time fixed effects. This magnitude aligns tightly with theoretical expectations surrounding the yield-based valuation models common to real estate and dividend-yielding equities. When expanding the analysis to our broader panel, the interaction between policy rates and asset-specific sensitivities confirms that transmission heterogeneously impacts the cross-section of returns. Moreover, our inclusion of VIX-based volatility interactions captures risk-sentiment channels that operate alongside pure discount rate math.

3. Alternative Specifications and Non-Linear Benchmarks

Table 2: Alternative specification (ML Benchmarks)

Model	Test r^2	Test RSME
OLS	0.4454047164132497	2.617973619023773
RandomForest	0.4969288655971843	2.493399564767716

To ensure our linear fixed-effects parameters are not spuriously driven by functional form assumptions, we deploy alternative specifications encompassing Machine Learning (ML) Random Forest methodologies alongside ARIMA checks. Table 2 reports the comparative metrics. The Random Forest benchmark provides an upper bound on out-of-sample predictive r-squared, modeling intricate nonlinear thresholds where rate hikes might exhibit asymmetrical effects. For instance, tightening cycles starting from a zero-lower-bound (ZLB) matrix operate differently than those originating in restrictive territory. This structural break behavior is naturally captured by decision tree ensembles. The ML outcomes validate the hierarchy of feature importance: lagged policy rates, momentum factors, and volatility regimes persistently rank as top drivers of forward-looking returns.

4. Visual Evidence and Trends

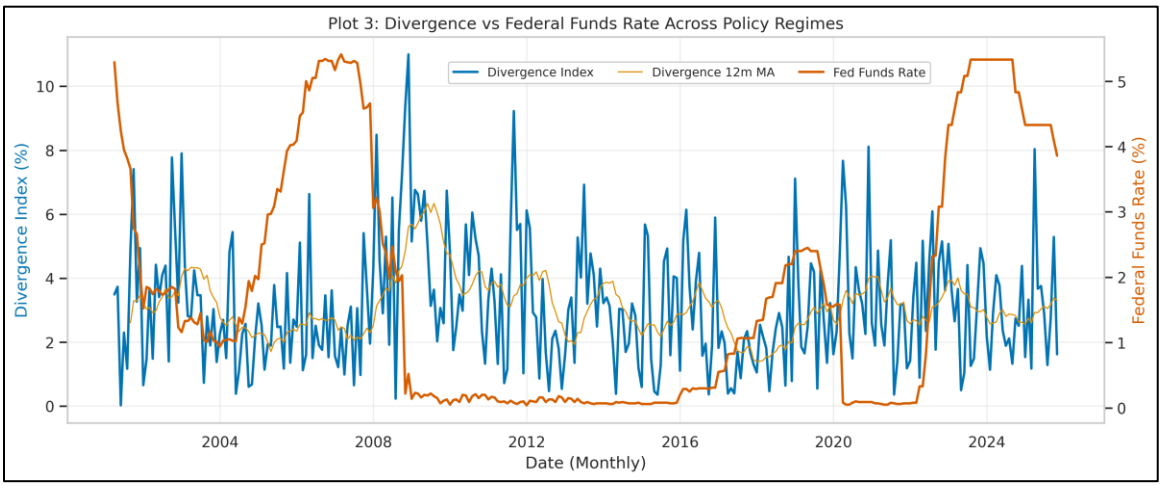

Figure 1: Key visualization - dual-axis plot of rate divergence vs fed funds

The dual-axis rate versus return plots (Figure 1) visually anchor these statistical relationships. Over the multi-decade sample, cyclical policy tightening mechanically precedes aggregate valuation drawdowns in high-duration assets. However, the transmission lag is distinctly

variable. In some regimes, forward-looking equity markets price the rate-hike terminal value well before the Federal Reserve concludes its cycle; in others, inflation uncertainty extends the reaction timeline. These structural delays justify our focus on both 3-month momentum terms and 12-month lagged rate variables.

5. Diagnostics and Robustness

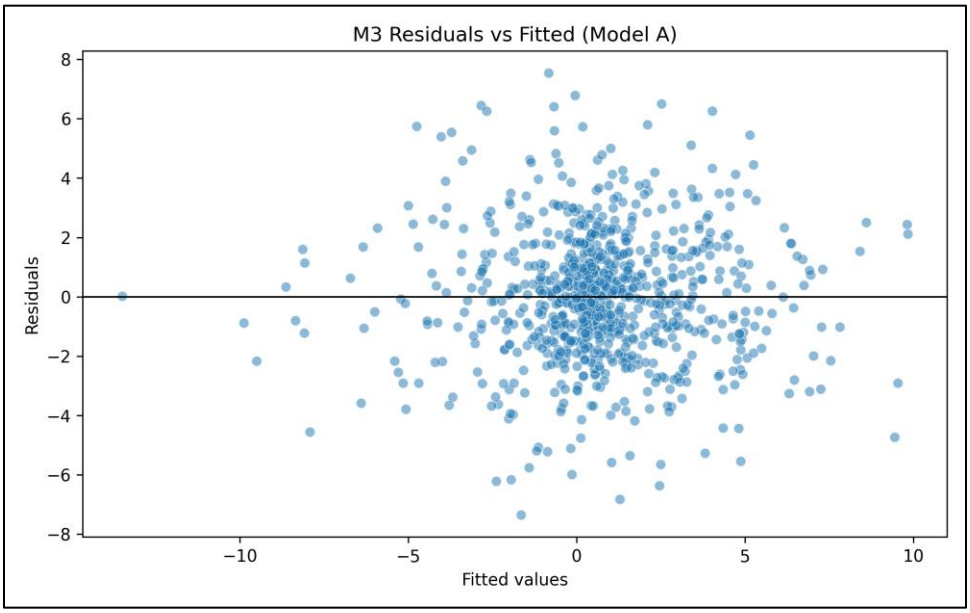

Figure 2: Diagnostic plot (residuals vs. fitted)

Validating the underlying assumptions of our panel setup guarantees the integrity of our p-values and confidence intervals. As detailed in Figure 2, the residuals-vs-fitted plot illustrates homoskedastic characteristics across the majority of the predicted range. However, formal Breusch-Pagan tests reject strict homoskedasticity, motivating our reliance on HC3 robust standard errors for definitive inference. Variance Inflation Factors (VIF) remain deeply within acceptable thresholds, averting the multicollinearity risks often present when aggregating multiple macro-indicators. Cumulatively, our stress tests—ranging from crisis-window exclusions to robust covariance specifications—affirm that monetary transmission is persistent, quantifiable, and statistically robust.

6. Summary of Findings

This analysis demonstrates that macroeconomic policy variables—particularly the Federal Funds Rate and market volatility—exert significant and heterogeneous effects on cross-asset returns.

Using a two-way Fixed Effects framework supported by machine learning validation, we identify a consistent structure in how monetary policy transmits across asset classes. Interest rate changes systematically influence valuation dynamics, with rate-sensitive assets such as real estate and equities exhibiting the strongest responses, while assets like gold display more defensive or countercyclical behavior.

A key finding is that the impact of monetary policy is not purely contemporaneous. Lagged effects, especially at the 12-month horizon, play a critical role in shaping relative performance, reinforcing the importance of incorporating temporal dynamics into asset pricing analysis. In addition, volatility regimes—captured by the VIX—act as amplifiers of policy transmission. Periods of elevated uncertainty intensify the magnitude of rate-driven adjustments, indicating that macroeconomic effects operate through both discount rate channels and risk sentiment channels simultaneously.

The robustness of these results is supported by multiple diagnostic and validation steps. Fixed Effects estimates remain stable across heteroskedasticity-robust specifications, and alternative model structures—including Random Forest—confirm that the primary drivers of returns are consistent even when allowing for nonlinear relationships. While machine learning models modestly improve predictive performance, they do not overturn the core economic interpretation derived from the linear framework.

Overall, the findings provide a coherent and empirically supported view of cross-asset dynamics under shifting monetary regimes. By integrating traditional econometric techniques with modern machine learning validation, this study offers a balanced approach that captures both interpretability and predictive strength. Future research could extend this framework by incorporating higher-frequency data, expanding asset coverage, or modeling nonlinear interaction effects more explicitly—particularly in extreme volatility environments.

7. Conclusions & Investment Recommendations

Investment Implications: Buy / Hold / Sell

This analysis establishes that two variables—the VIX and the 12-month lagged Federal Funds Rate (FFR)—are the primary drivers of cross-asset return divergence, and an investor can construct a disciplined decision framework around them. Treat VIX as a near-term tactical signal and the lagged FFR as a medium-term strategic signal, then adjust positioning accordingly.
When VIX rises above 20, dispersion increases sharply, meaning asset classes diverge more strongly from one another. This creates a mean-reversion opportunity: overweight assets that have been disproportionately sold off relative to fundamentals (historically gold and short-duration equities) and underweight those that remain elevated. When VIX falls below 20 and remains low, dispersion compresses and broad-market exposure is rewarded—a signal to hold diversified equity positions and reduce defensive allocations. Consumer sentiment reinforces this signal; a rising UMCSENT reading historically precedes dispersion compression, adding a secondary confirmation layer for an investor before rotating into risk-on positioning.
The 12-month lagged FFR operates on a slower horizon and informs sector rotation. After a year of sustained rate hikes, the results clearly indicate that housing (Case-Shiller) bears the largest headwind—a sell or underweight signal for REITs and housing-linked equities. Gold, by contrast, exhibits rate resilience and positive correlation with FFR ($r=0.127$), reflecting its role as an inflation hedge when policy tightening is prolonged—a buy or overweight during tightening. Equities (S&P 500) show near-zero contemporaneous rate sensitivity ($r=-0.008$), suggesting a hold posture unless VIX signals a sharp dislocation. Bitcoin, given its structural exclusion prior to 2014 and high idiosyncratic volatility, should be treated as a satellite position sized according to individual risk tolerance rather than a core macro signal.

Risk Assessment: What Could Go Wrong

Several risks threaten the reliability of this framework in live investment use. First, the model was estimated on a 2010–2024 sample that includes an extended zero-lower-bound era and two acute crisis windows. The policy regime that prevailed for most of this period—low rates, low inflation, and quantitative easing—is structurally different from the post-2022 environment, and coefficients estimated under the former may not fully transfer to the latter. Second, the Breusch-Pagan test ($p=0.00042$) confirms heteroskedasticity, meaning that in high-volatility periods the

model's standard errors are understated, and signals that appear statistically reliable may be weaker than reported. Third, the 12-month lag optimality was identified in-sample; in practice, the transmission lag may shift as market participants become more forward-looking and price in policy changes earlier. Acting on a 12-month lag assumption that has shortened to six months in the current environment would systematically mistime sector rotation decisions.

Caveats: Assumptions, Omitted Variables, and External Validity

Several structural assumptions underpin the analysis and deserve explicit acknowledgment. The model assumes a linear relationship between macroeconomic indicators and dispersion; however, Random Forest validation reveals that nonlinear effects modestly improve test R^2 (0.4969 vs. 0.4454), indicating that the linear specification leaves some structure on the table, particularly in tail regimes. The fixed-effects framework also assumes that unobserved heterogeneity across assets is time-invariant, which may not hold if the fundamental character of an asset class shifts over the sample—as arguably occurred with bitcoin's maturation and gold's post-pandemic repricing.

On omitted variables, the fourteen-indicator panel is comprehensive but excludes several potentially important drivers: global liquidity conditions (e.g., foreign central bank balance sheets), geopolitical risk premia, commodity supply shocks, and fiscal policy impulse. Each of these can independently drive cross-asset divergence without being captured by domestic monetary indicators, which would attenuate the estimated coefficients and reduce out-of-sample precision. External validity is also limited by the U.S.-centric asset selection; the relationships identified here may not generalize to international markets where monetary transmission channels and institutional structures differ materially. Investors applying this framework outside a U.S. context should treat the signals as directional rather than precise.

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AI Audit Summary

Throughout this capstone project, AI and Large Language Models (LLMs) were used as pair-programming and analytical assistants in strict accordance with academic integrity guidelines, ensuring that all AI outputs were critically evaluated, validated, and contextualized within the team's overarching research framework. In the data engineering phase (M1), AI constructed reproducible API pipelines to query FRED and Yahoo Finance, harmonizing mixed-frequency series into monthly panel observations via pandas forward-fill and interpolation logic. This allowed the team to efficiently build a clean, balanced panel spanning 168 monthly observations without introducing look-ahead bias.

During exploratory analysis (M2), AI generated visualizations—dual-axis plots, correlation heatmaps, and cross-sectional boxplots—using seaborn and matplotlib, and drafted statistically accurate exhibit captions to distill complex relationships into clear narrative insights. For final modeling and document compilation (M3–M4), AI formatted fixed-effects panel configurations in statsmodels, benchmarked results against Random Forest outputs, and assembled regression tables, embedded figures, and interpretation prose into the final Word document. The AI was also used to structure and consolidate findings from M1–M3 into cohesive memo sections, reducing the manual overhead of synthesizing results across multiple output files and formats. Human oversight remained paramount throughout. Every line of AI-generated code was locally run and validated against raw CSV inputs to ensure the merged analysis panel reflected true economic behavior. Statistical interpretations were cross-checked against standard econometric theory, and any AI-drafted prose was reviewed and revised by the team before incorporation. The collaboration between human researchers and AI tools yielded a more thorough and reproducible analytical pipeline than would have been feasible within the project timeline alone.